H2 Math Summary

Example Questions

Maximus Pung

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1 Systems of Linear Equations

1.1 Three different blends of coffee beans – Blend A, Blend B and Blend C – are sold at both M Café and I Café. The usual selling prices of the blends are the same at both cafés. The total cost of buying one package each of Blend A, Blend B and Blend C is \$176. During a holiday sale the two cafés offered the following discounts:

Café	Discounts given for each package			Total price after the discount
	Blend A	Blend B	Blend C	
M Café	15%	10%	5%	\$161.80
I Café	7%	5%	2%	\$169.56

(i) Find the usual selling price of each package of Blend A, Blend B and Blend C coffee beans respectively. [3]

[MI 2025 Prelim P1 Q4]

Solution		Comments
(i) **[3]**	Let the usual prices be \$$a$, \$$b$, \$$c$. $a+b+c=176 \quad ---(1)$ $0.85a+0.90b+0.95c=161.80 \quad ---(2)$ $0.93a+0.95b+0.98c=169.56 \quad ---(3)$ Using GC, $a=32,\ b=44,\ c=100$ $\therefore$ Blend A: \$32, Blend B: \$44, Blend C: \$100	

1.2 A function f is defined by $\mathrm{f}(x)=ax^3+bx^2+cx+d$ where a, b, c and d are constants. The graph of $y=\mathrm{f}(x)$ has a turning point at $x=-1$ and passes through the points $(2, 10)$ and $(-2, -2)$. Given also that $\int_{-2}^{2} \mathrm{f}(x)\,\mathrm{d}x=16$, find the values of a, b, c and d. [4]

[SAJC 2025 Prelim P1 Q1]

Solution		**Comments**
[4]	$\mathrm{f}'(x)=3ax^2+2bx+c$ $\mathrm{f}'(-1)=0$ $3a-2b+c=0 \quad ---(1)$ $\mathrm{f}(2)=10$ $8a+4b+2c+d=10 \quad ---(2)$ $\mathrm{f}(-2)=-2$ $-8a+4b-2c+d=-2 \quad ---(3)$ $\int_{-2}^{2} (ax^3+bx^2+cx+d)\,\mathrm{d}x=16$ $\left[\frac{a}{4}x^4+\frac{b}{3}x^3+\frac{c}{2}x^2+dx\right]_{-2}^{2}=16$ $(4a+\frac{8}{3}b+2c+2d)-(4a-\frac{8}{3}b+2c-2d)=16$ $\frac{16}{3}b+4d=16 \quad ---(4)$ Using GC, $a=3,\ b=0,\ c=-9,\ d=4$	

Two linear equations are independent if and only if neither equation is a linear combination of the other.

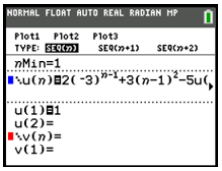
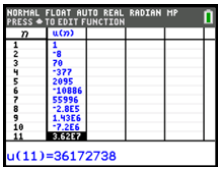
1.3 A sequence of real numbers, x_n, satisfies the recurrence relation

$$x_{n+1} = a(-3)^n + bn^2 + cx_n, \text{ for } n, a, b, c \in \mathbb{Z} \text{ and } n \geqslant 1.$$

Given that $x_1 = 1$, $x_2 = -8$, $x_3 = 70$, and $x_4 = -377$, find the values of a, b and c. [3]

Hence, find the values of x_{10} and x_{11}. [2]

[HCI 2025 Prelim P1 Q2]

Solution		**Comments**
[2]	$n = 1$ $x_2 = a(-3)^1 + b(1)^2 + cx_1$ $-8 = -3a + b + c \quad ---(1)$ $n = 2$ $x_3 = a(-3)^2 + b(2)^2 + cx_2$ $70 = 9a + 4b - 8c \quad ---(2)$ $n = 3$ $x_4 = a(-3)^3 + b(3)^2 + cx_3$ $-377 = -27a + 9b + 70c \quad ---(3)$ Using GC, $a = 2,\ b = 3,\ c = -5$ $x_{n+1} = 2(-3)^n + 3n^2 - 5x_n$ Using GC, $x_{10} = -7\,210\,868$ $x_{11} = 36\,172\,738$   	

1.4 The lines l_1 and l_2 have equations

$$l_1: \mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix},$$

$$l_2: \mathbf{r} = \begin{pmatrix} 0 \\ 5 \\ -8 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ -2 \\ 3 \end{pmatrix},$$

where λ and μ are parameters.

(i) Show that l_1 and l_2 are skew lines. [2]

[SAJC 2025 Prelim P1 Q3]

Solution		**Comments**
(i) **[2]**	$1-2\lambda = 2\mu$ $---(1)$ $2+\lambda = 5-2\mu$ $---(2)$ $2\lambda = -8+3\mu$ $---(3)$ Solve (1) and (2): $\lambda = -2,\ \mu = \frac{5}{2}$ Sub into (3): $\text{LHS} = 2(-2) = -4$ $\text{RHS} = -8+3\left(\frac{5}{2}\right) = -\frac{1}{2}$ $\text{LHS} \neq \text{RHS},$ no solution $\therefore l_1$ and l_2 do not intersect $\because \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix} \neq k \begin{pmatrix} 2 \\ -2 \\ 3 \end{pmatrix}$ for any $k \in \mathbb{R}$ $\therefore l_1$ and l_2 are not parallel $\therefore l_1$ and l_2 are skew (shown)	

1.5 A movie theatre supports a charity event by providing discounted movie tickets at the following prices based on age group.

Age group	Discounted ticket price
$\leqslant 12$ years	\$6
13 to 49 years	\$9
$\geqslant 50$ years (senior citizen)	\$4

The event organiser plans to spend \$700 to sponsor 100 participants of which at least 35 of them are senior citizens. Find the possible number of participants for each age group. [4]

[TJC 2022 Prelim P2 Q1]

Solution		**Comments**
[4]	Let x, y, z be the number of participants aged $\leqslant 12$, 13 to 49 and $\geqslant 50$ years respectively. $\left.\begin{aligned} x+y+z&=100 \quad ---(1)\\ 6x+9y+4z&=700 \quad ---(2)\end{aligned}\right\} \Rightarrow$ Using GC, $x=\frac{200}{3}-\frac{5}{3}z,\ y=\frac{100}{3}+\frac{2}{3}z$ $z \geqslant 35$ $\because x \geqslant 0$ $\frac{200}{3}-\frac{5}{3}z \geqslant 0$ $z \leqslant 40$ $\therefore 35 \leqslant z \leqslant 40$ x, y, z are non-negative integers. Using GC table: $z=37 \Rightarrow x=5,\ y=58$ $z=40 \Rightarrow x=0,\ y=60$ $\therefore (x,y,z)=(5,58,37)$ or $(0,60,40)$	

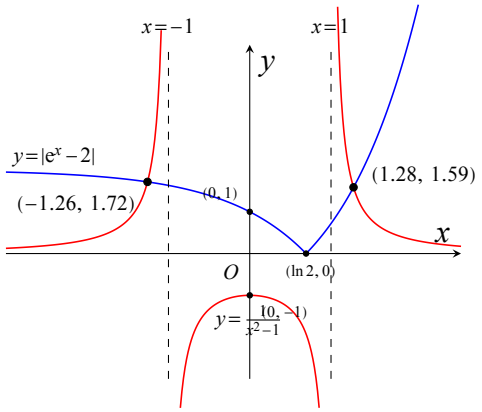
2 Inequalities

2.1 **(a)** Solve the inequality $|e^x - 2| > \dfrac{1}{x^2 - 1}$ using a graphical approach. [3]

(b) Without the use of a graphing calculator, solve the inequality $\dfrac{x+2}{x-1} \geqslant \dfrac{2}{x-3}$. [3]

[RVHS 2026 Prelim P1 Q2]

Solution	Comments
(a) **[3]**	



Using GC, intersections at $x = -1.2581$, $x = 1.2770$

$\therefore x < -1.26$ or $-1 < x < 1$ or $x > 1.28$

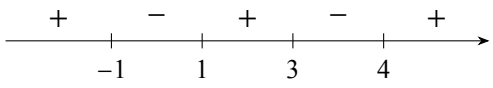
(b) **[3]**

$$\frac{x+2}{x-1} - \frac{2}{x-3} \geqslant 0$$

$$\frac{(x+2)(x-3) - 2(x-1)}{(x-1)(x-3)} \geqslant 0$$

$$\frac{x^2 - 3x - 4}{(x-1)(x-3)} \geqslant 0$$

$$\frac{(x-4)(x+1)}{(x-1)(x-3)} \geqslant 0$$



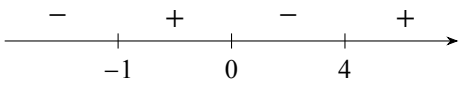
$\therefore x \leqslant -1$ or $1 < x < 3$ or $x \geqslant 4$

2.2 Use an algebraic method to solve the inequalities

$$\frac{x(5x+9)}{x^2+2x+5} > x \geqslant 3x-6.$$

[4]

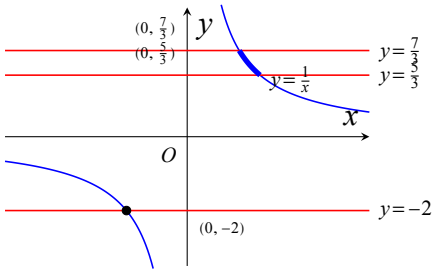
[ACJC 2026 Prelim P1 Q1]

Solution		Comments
[4]	$x \geqslant 3x-6$ $x \leqslant 3 \quad ---(1)$ $x^2+2x+5=(x+1)^2+4>0$ for all real x $\frac{x(5x+9)}{x^2+2x+5} > x$ $5x^2+9x > x(x^2+2x+5) \quad (\because x^2+2x+5>0)$ $x^3-3x^2-4x<0$ $x(x-4)(x+1)<0$  $x<-1$ or $0<x<4 \quad ---(2)$ $(1) \cap (2)$: $\therefore x<-1$ or $0<x \leqslant 3$	

2.3 **(i)** Solve the inequality $3\,|x^2-4| \leqslant |x+2|$. [4]

(ii) Hence, solve $\dfrac{3}{x^2}\,|1-4x^2| \leqslant \left|\dfrac{1+2x}{x}\right|$. [2]

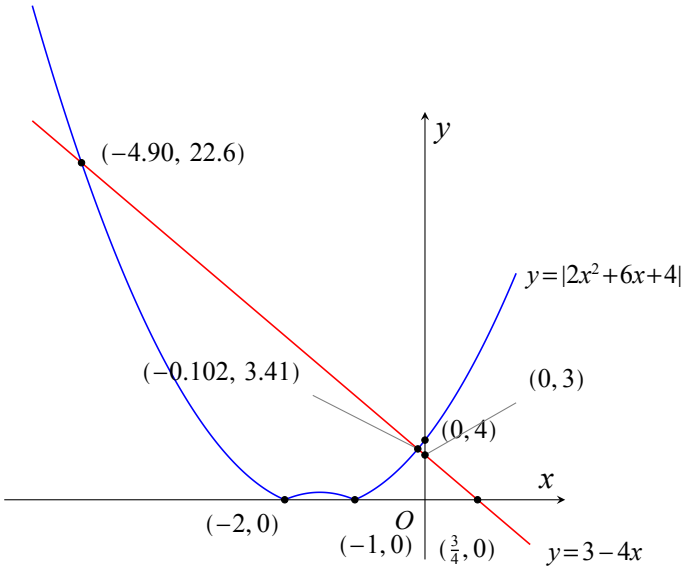
[SAJC 2025 Prelim P2 Q1]

Solution		Comments
(i) **[4]**	$3\|x-2\|\|x+2\| \leqslant \|x+2\|$ $x=-2:\quad 0 \leqslant 0$ (true) $x \neq -2:\ \|x+2\|>0$ $3\|x-2\| \leqslant 1$ $-\frac{1}{3} \leqslant x-2 \leqslant \frac{1}{3}$ $\frac{5}{3} \leqslant x \leqslant \frac{7}{3}$ $\therefore x=-2$ or $\frac{5}{3} \leqslant x \leqslant \frac{7}{3}$	
(ii) **[2]**	$\dfrac{3}{x^2}\,\|1-4x^2\| = 3\left\|\dfrac{1}{x^2}-4\right\| \quad (\because x^2>0)$ $\left\|\dfrac{1+2x}{x}\right\| = \left\|\dfrac{1}{x}+2\right\|$ $3\left\|\left(\dfrac{1}{x}\right)^2-4\right\| \leqslant \left\|\dfrac{1}{x}+2\right\| \quad (x \to \frac{1}{x})$ $\dfrac{1}{x}=-2$ or $\dfrac{5}{3} \leqslant \dfrac{1}{x} \leqslant \dfrac{7}{3}$  $\therefore x=-\frac{1}{2}$ or $\frac{3}{7} \leqslant x \leqslant \frac{3}{5}$	

2.4 **(i)** On the same axes, sketch the curves with equations $y=|2x^2+6x+4|$ and $y=3-4x$, indicating any intercepts with the axes and points of intersection. Hence solve the inequality $3-4x<|2x^2+6x+4|$. [4]

(ii) Find the exact area bounded by the graphs of $y=3-4x$, $y=|2x^2+6x+4|$, $x=-3$ and $x=-1$. [3]

[SAJC 2019 Prelim P1 Q2]

Solution		**Comments**
(i) **[4]**		
	$3-4x=2x^2+6x+4$ $2x^2+10x+1=0$ $x=\dfrac{-5\pm\sqrt{23}}{2}$ $\therefore x<\dfrac{-5-\sqrt{23}}{2}$ or $x>\dfrac{-5+\sqrt{23}}{2}$	
(ii) **[3]**	$\text{Area}=\int_{-3}^{-1}(3-4x-\lvert2x^2+6x+4\rvert)\,dx$ $=\int_{-3}^{-2}(3-4x-2x^2-6x-4)\,dx+\int_{-2}^{-1}(3-4x+2x^2+6x+4)\,dx$ $=\int_{-3}^{-2}(-2x^2-10x-1)\,dx+\int_{-2}^{-1}(2x^2+2x+7)\,dx$ $=\left[-\frac{2}{3}x^3-5x^2-x\right]_{-3}^{-2}+\left[\frac{2}{3}x^3+x^2+7x\right]_{-2}^{-1}$ $=\left(\frac{16}{3}-20+2\right)-(18-45+3)+\left(-\frac{2}{3}+1-7\right)-\left(-\frac{16}{3}+4-14\right)$ $=20$	

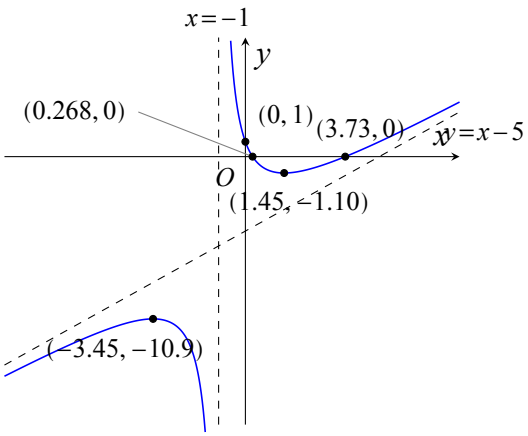
3 Graphing Techniques

3.1 A curve C has equation $y=\dfrac{x^2-4x+1}{x+1}$, $x\neq -1$.

(a) Prove, using an algebraic method, that $y=\dfrac{x^2-4x+1}{x+1}$ cannot lie between two certain values, to be determined in exact form. [3]

(b) Sketch C, stating the equations of any asymptotes, and the coordinates of any turning points and any axial intercepts. [4]

[VJC 2025 Promo Q9(a), (b)]

Solution		Comments
(a) **[3]**	$y=\dfrac{x^2-4x+1}{x+1}$ $yx+y=x^2-4x+1$ $x^2+(-y-4)x+(1-y)=0$ No real solution for x: $(-y-4)^2-4(1-y)<0$ $y^2+8y+16-4+4y<0$ $y^2+12y+12<0$ $(y+6)^2-24<0$ $\left(y+6-\sqrt{24}\right)\left(y+6+\sqrt{24}\right)<0$ $-6-2\sqrt{6}<y<-6+2\sqrt{6}$ $\therefore y$ cannot lie between $-6-2\sqrt{6}$ and $-6+2\sqrt{6}$	
(b) **[4]**	$y=x-5+\dfrac{6}{x+1}$ Asymptotes: $x=-1,\ y=x-5$ $x=0$: $y=1$ $y=0$: $x^2-4x+1=0 \Rightarrow x=0.268,\ 3.73$ Turning points: $(1.45,-1.10),\ (-3.45,-10.9)$  	

3.2 A curve C has equation $y=ax+b+\dfrac{2b-a}{x+3}$, where a and b are constants such that $a>0$, $b\neq\dfrac{1}{2}a$ and $x\neq -3$.

(a) If C has no stationary points, use differentiation to find the relationship between a and b. [3]

(b) It is now given that $b=a$.

(i) Find the x-coordinates of the turning points. [2]

(ii) Hence sketch C, stating the equations of any asymptotes and the coordinates of the axial intercepts and turning points. [3]

(iii) Verify that the point of intersection of the two asymptotes of C lies on the line $y=kx+3k-2a$.

Hence, using the graph in part (b)(ii), find the range of values of k in terms of a, such that the equation $x(k-a)+3k-2a-b-\dfrac{2b-a}{x+3}=0$ has real roots. [2]

[SAJC 2025 Promo Q7]

Solution		Comments
(a) **[3]**	$\dfrac{dy}{dx}=a-\dfrac{2b-a}{(x+3)^2}$ $\dfrac{dy}{dx}=0:\quad a-\dfrac{2b-a}{(x+3)^2}=0$ $(x+3)^2=\dfrac{2b-a}{a}\qquad ---(1)$ No stationary points $\Rightarrow \dfrac{2b-a}{a}<0$ Since $a>0,\quad 2b-a<0$ $\therefore a>2b$	
(b)(i) **[2]**	$b=a:\quad (x+3)^2=\dfrac{2a-a}{a}=1$ $x+3=1$ or -1 $x=-2$ or -4	

(b)(ii)
[3]

$$y = ax + a + \frac{a}{x+3}$$

Asymptotes: $x = -3,\ y = ax + a$

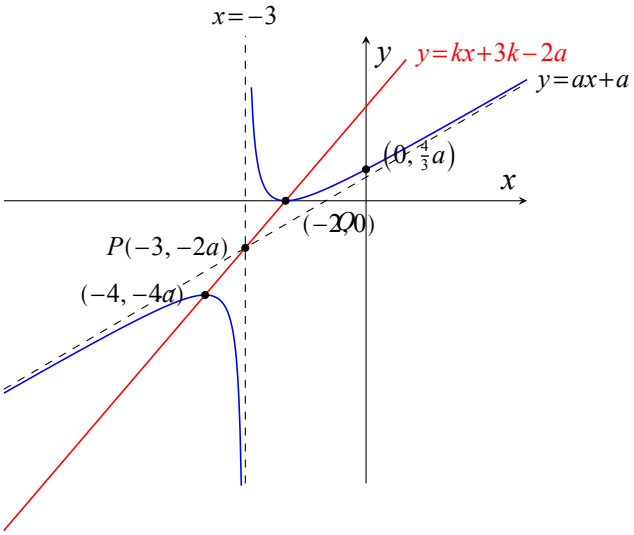
$x = -2 : y = 0; \quad x = -4 : y = -4a$

Turning points: $(-2, 0),\ (-4, -4a)$

$$x = 0 : y = \frac{4}{3}a$$

$$y = 0 : ax + a + \frac{a}{x+3} = 0 \Rightarrow (x+1)(x+3) + 1 = 0$$

$$x^2 + 4x + 4 = 0 \Rightarrow (x+2)^2 = 0 \Rightarrow x = -2$$



(b)(iii)
[2]

$x = -3$ and $y = ax + a$: $\quad y = -3a + a = -2a$

Intersection of asymptotes: $P(-3, -2a)$

$x = -3$: $\quad \text{RHS} = k(-3) + 3k - 2a = -2a = \text{LHS}$

$\therefore P$ lies on the line $y = kx + 3k - 2a$

$$x(k-a) + 3k - 2a - b - \frac{2b-a}{x+3} = 0$$

$$kx + 3k - 2a = ax + b + \frac{2b-a}{x+3}$$

Line through P meets $C \Rightarrow k > a$

3.3 A curve C has equation $\mathrm{f}(x)=\dfrac{3x^2+6x+k}{x+2}$, where $x>-2$ and k is a non-zero real constant.
It is given that the function f is strictly increasing.

(a) Find the range of values of k. [3]

(b) Sketch the curve C, giving the coordinates of any points where C crosses the x- and y-axes and the equations of any asymptotes. [3]

(c) By adding a suitable curve to the graph of the curve C in (b), deduce the number of distinct real root(s) of the equation $(3x^2+6x+k)^2=(x^2+1)\,(9x^2+36x+36)$. [3]

[DHS 2025 Promo Q8]

Solution		Comments
(a) **[3]**	$\mathrm{f}(x)=\dfrac{3x^2+6x+k}{x+2}=3x+\dfrac{k}{x+2}$	
	$\mathrm{f}'(x)=3-\dfrac{k}{(x+2)^2}=\dfrac{3x^2+12x+12-k}{(x+2)^2}$	
	f strictly increasing $\Rightarrow$ $\mathrm{f}'(x)>0$	
	$3x^2+12x+12-k>0 \quad \because (x+2)^2>0$ for $x>-2$	
	$3(x+2)^2-k>0$	
	Since $3(x+2)^2-k>-k$ for all $x>-2$, $\quad -k\geqslant 0$	
	$k\neq 0 \Rightarrow k<0$	
(b) **[3]**	$y=3x+\dfrac{k}{x+2}$	
	Asymptotes: $y=3x,\ x=-2$	
	$x=0:\ y=\dfrac{k}{2}$	
	$y=0:\ 3x^2+6x+k=0 \Rightarrow x=\dfrac{-6\pm\sqrt{36-12k}}{6}=-1\pm\dfrac{\sqrt{9-3k}}{3}$	
	$x>-2 \Rightarrow x=-1+\dfrac{\sqrt{9-3k}}{3}$	

(c)
[3]

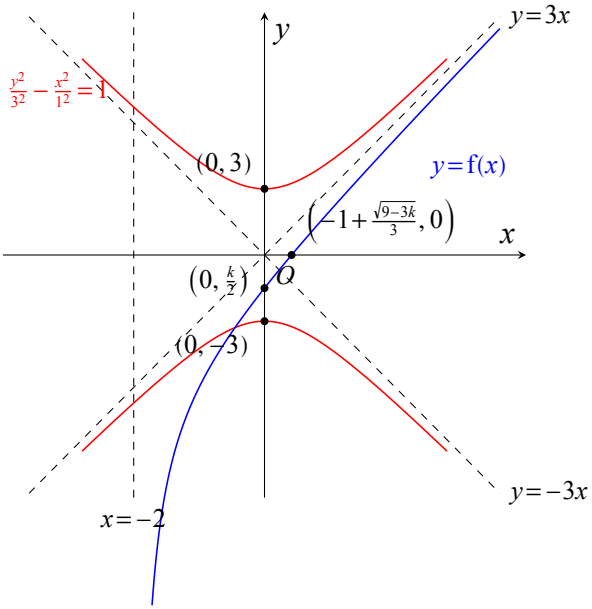
$$(3x^2+6x+k)^2=(x^2+1)\,(9x^2+36x+36) \quad ---(*)$$

$$(3x^2+6x+k)^2=9\,(x^2+1)\,(x+2)^2$$

$$\left(\frac{3x^2+6x+k}{x+2}\right)^2-9x^2=9$$

Draw the hyperbola $\dfrac{y^2}{3^2}-\dfrac{x^2}{1^2}=1$

Asymptotes $y=\pm 3x$, vertices $(0,\pm 3)$



$y=\mathrm{f}(x)$ and $\dfrac{y^2}{3^2}-\dfrac{x^2}{1^2}=1$ have one intersection point

$\therefore$ $(*)$ has one real root

3.4 The curve C is defined by the parametric equations $x=6t-5$ and $y=2t^2+1$, where $t \geqslant \frac{5}{6}$.

(a) Sketch the curve C, labelling the exact coordinates of any axial intercept(s). [1]

(b) Find the equation of the normal to C at the point P where $t=1$. [2]

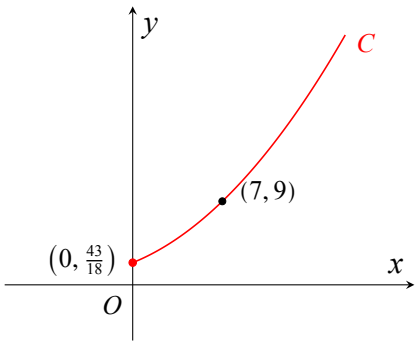
(c) Find the acute angle between the normal to C at the point P where $t=1$ and the tangent to C at the point M where $t=3$. [2]

(d) The curve C is translated 5 units in the positive x-direction and translated 3 units in the negative y-direction, to form the curve D. Find the equation of D in parametric form. [2]

The curve E is defined by the parametric equations $x=7u$ and $y=\frac{9}{u}$, where $u \neq 0$.

(e) Show that at the point of intersection of the curves C and E, $6t^3-5t^2+3t-34=0$. Deduce that there is only one point of intersection and find the coordinates of this point. [4]

[RVHS 2025 Prelim P1 Q12]

Solution		**Comments**
(a) **[1]**	$t=\frac{5}{6}$: $x=6\left(\frac{5}{6}\right)-5=0$, $y=2\left(\frac{5}{6}\right)^2+1=\frac{43}{18}$ Endpoint (y-intercept): $\left(0, \frac{43}{18}\right)$  	

(b) **[2]**

$$\frac{dx}{dt}=6, \quad \frac{dy}{dt}=4t$$

$$\frac{dy}{dx}=\frac{4t}{6}=\frac{2}{3}t$$

$t=1$: $\quad \frac{dy}{dx}=\frac{2}{3} \Rightarrow$ gradient of normal $=-\frac{3}{2}$

$t=1$: $\quad x=1,\ y=3$

$$y-3=-\frac{3}{2}(x-1)$$

$$y=-\frac{3}{2}x+\frac{9}{2}$$

(c) **[2]**

$t=3$: $\quad$ gradient of tangent $=\frac{2}{3}(3)=2$

$$\theta_1+\theta_2=\tan^{-1}(2)-\tan^{-1}\left(-\frac{3}{2}\right)=2.089942 \text{ or } 119.7^\circ$$

$$\text{Acute angle}=\pi-2.089942 \text{ or } 180^\circ-119.7^\circ$$

$$=1.05 \text{ or } 60.3^\circ$$

(d) **[2]**

$x-5=6t-5 \Rightarrow x=6t$

$y+3=2t^2+1 \Rightarrow y=2t^2-2$

D: $\quad x=6t,\ y=2t^2-2$

(e) **[4]**

$$6t-5=7u \Rightarrow u=\frac{6t-5}{7} \quad ---(1)$$

$$2t^2+1=\frac{9}{u} \quad ---(2)$$

Sub (1) into (2): $2t^2+1=\frac{63}{6t-5}$

$$(2t^2+1)(6t-5)=63$$

$$12t^3-10t^2+6t-68=0$$

$$6t^3-5t^2+3t-34=0 \quad \text{(shown)}$$

$$(t-2)(6t^2+7t+17)=0$$

$6t^2+7t+17$: $b^2-4ac=7^2-4(6)(17)=-359<0 \Rightarrow$ no real roots

$\therefore t=2$ only $\Rightarrow$ only one point of intersection

$t=2$: $\quad x=6(2)-5=7,\ y=2(2)^2+1=9$

Point of intersection: $(7, 9)$

3.5 The parametric equations of the curve C are

$$x=\tan\theta-1 \quad \text{and} \quad y=2\sec\theta, \quad \text{where } -\frac{\pi}{2}<\theta<\frac{\pi}{2}.$$

(a) Find $\frac{\mathrm{d}y}{\mathrm{d}x}$ in terms of θ, leaving your answer in a single trigonometric function. [2]

(b) Find the equation of the tangent to C at the point where it cuts the y-axis, leaving your answer in exact form. [3]

(c) Find the Cartesian equation of C. [1]

(d) Find the equations of the asymptotes and state the coordinates of any turning point(s). [3]

(e) The region bounded by the tangent found in part (b), the curve C and the line $x=-1$ is rotated completely about the x-axis. Calculate the volume obtained. [3]

[HCI 2024 Promo Q12]

Solution		**Comments**
(a) **[2]**	$\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{\mathrm{d}y}{\mathrm{d}\theta}\div\frac{\mathrm{d}x}{\mathrm{d}\theta}=\frac{2\sec\theta\tan\theta}{\sec^2\theta}=2\sin\theta$	
(b) **[3]**	$x=0$: $\tan\theta-1=0 \Rightarrow \tan\theta=1$ $\theta=\frac{\pi}{4}$ since $-\frac{\pi}{2}<\theta<\frac{\pi}{2}$ $\theta=\frac{\pi}{4}$: $y=\frac{2}{\cos\frac{\pi}{4}}=2\sqrt{2}$, $\frac{\mathrm{d}y}{\mathrm{d}x}=2\sin\frac{\pi}{4}=\sqrt{2}$ Tangent: $y=\sqrt{2}x+2\sqrt{2}$	
(c) **[1]**	$\tan^2\theta+1=\sec^2\theta$ $(x+1)^2+1=\left(\frac{y}{2}\right)^2$ $-\frac{\pi}{2}<\theta<\frac{\pi}{2} \Rightarrow 0<\cos\theta\leqslant 1 \Rightarrow \sec\theta\geqslant 1$ $\therefore (x+1)^2+1=\left(\frac{y}{2}\right)^2, \quad y\geqslant 2$	

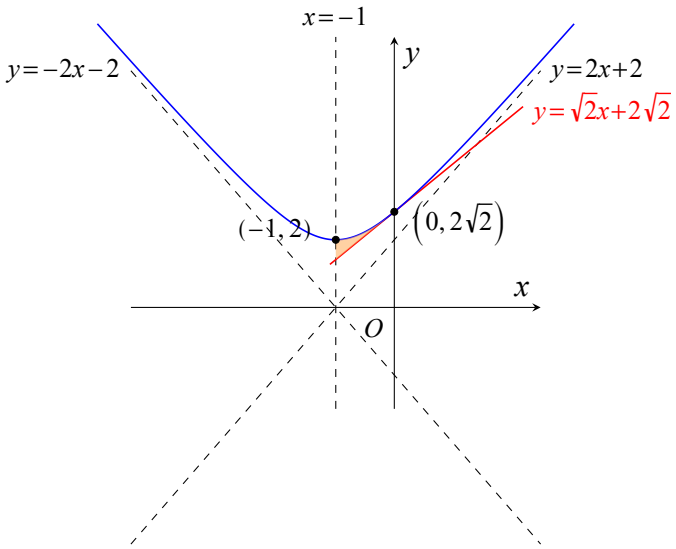
(d)
[3]

Asymptotes: $\left(\frac{y}{2}\right)^2 = (x+1)^2$

$$\frac{y}{2} = \pm(x+1)$$

$y = 2x+2$ or $y = -2x-2$

Centre $(-1, 0)$, $y \geqslant 2 \Rightarrow$ turning point $(-1, 2)$



(e)
[3]

$$\frac{y^2}{4} = (x+1)^2 + 1$$

$$y^2 = 4(x+1)^2 + 4$$

$$\text{Volume} = \pi \int_{-1}^{0} \left[4\left((x+1)^2+1\right) - \left(\sqrt{2}x + 2\sqrt{2}\right)^2 \right] dx$$

$$= 2.09 \text{ units}^3$$

4 Graphing Transformations

4.1 A curve with equation $y=\mathrm{h}(x)$ undergoes the following sequence of transformations.

A: A translation of 2 units in the positive direction of the x-axis.

B: A scaling parallel to the x-axis by factor of 3.

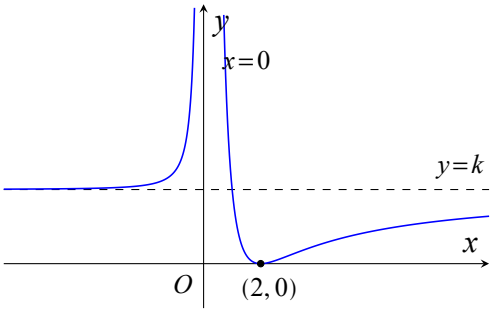
C: A reflection in the y-axis.

The resultant curve has equation $y=-\dfrac{x^2}{6x+45}$. Find $\mathrm{h}(x)$. [3]

[NJC 2024 Promo Q4(b)]

Solution		Comments
[3] **(b)**	$y=-\dfrac{x^2}{6x+45}$ C^{-1}: reflection in the y-axis $(x\to -x)$ $y=-\dfrac{(-x)^2}{6(-x)+45}=-\dfrac{x^2}{-6x+45}$ B^{-1}: scaling parallel to the x-axis by factor $\frac{1}{3}$ $(x\to 3x)$ $y=-\dfrac{(3x)^2}{-6(3x)+45}=-\dfrac{9x^2}{-18x+45}=-\dfrac{x^2}{-2x+5}$ A^{-1}: translation of 2 units in the negative x-direction $(x\to x+2)$ $y=-\dfrac{(x+2)^2}{-2(x+2)+5}=-\dfrac{(x+2)^2}{-2x+1}$ $\mathrm{h}(x)=\dfrac{(x+2)^2}{2x-1}$	

4.2 The diagram below shows the graph of $y=\mathrm{f}(2x)$. The graph has a turning point at $(2,0)$, and asymptotes with equations $x=0$ and $y=k$.



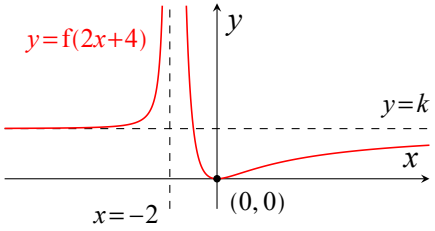
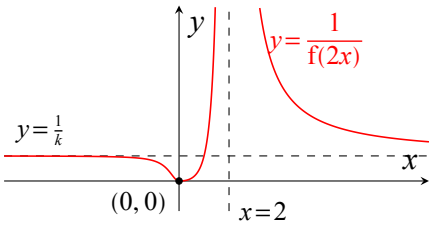
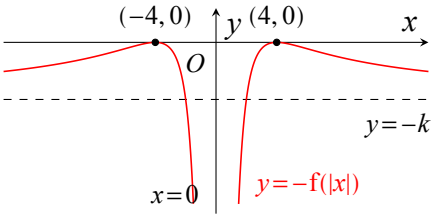
(a) State a single transformation that will transform the graph of $y=\mathrm{f}(2x)$ onto the graph of $y=\mathrm{f}(2x+4)$. Hence sketch the graph of $y=\mathrm{f}(2x+4)$. [3]

On separate clearly labelled diagrams, sketch the graphs of

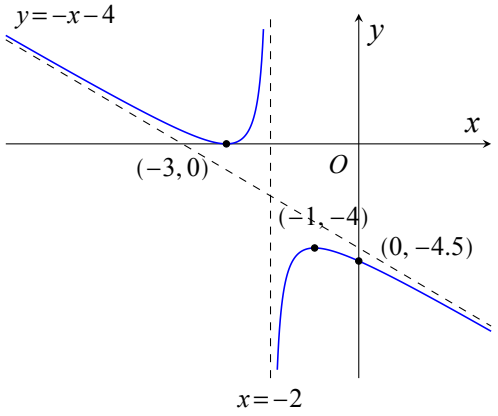
(b) $y=\dfrac{1}{\mathrm{f}(2x)}$, [3]

(c) $y=-\mathrm{f}(|x|)$. [3]

[EJC 2025 Prelim P1 Q4]

Solution		Comments
(a) **[3]**		
(b) **[3]**		
(c) **[3]**		

4.3



The diagram shows the curve $y = \mathrm{f}(x)$. The curve crosses the x-axis at its minimum point $(-3, 0)$. It has a maximum turning point $(-1, -4)$ and crosses the y-axis at the point $(0, -4.5)$. The asymptotes of the curve are $x = -2$ and $y = -x - 4$.

On separate diagrams, sketch the graphs of

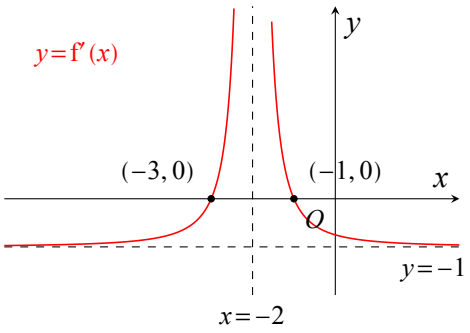
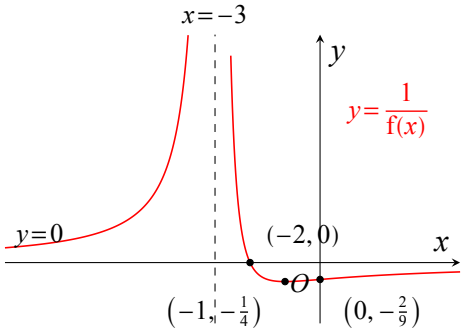
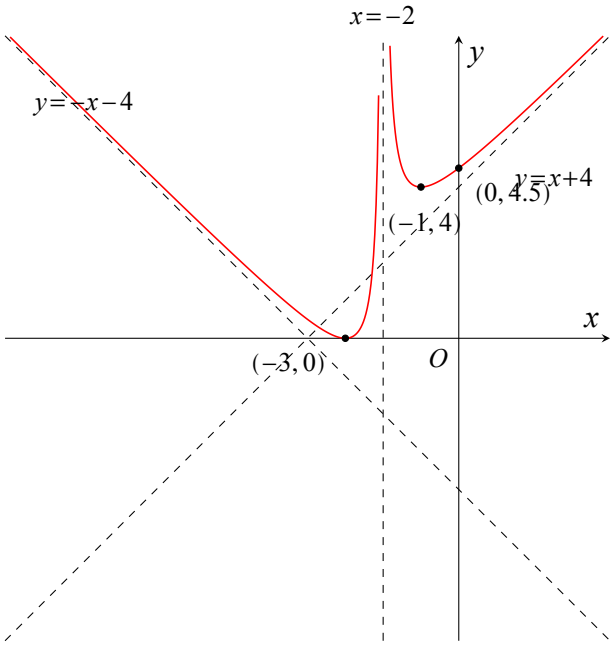
(a) $y = \mathrm{f}'(x)$, [3]

(b) $y = \dfrac{1}{\mathrm{f}(x)}$, [3]

(c) $y = |\mathrm{f}(x)|$, [3]

indicating the equations of any asymptotes, turning points and intersections with the axes.

[CJC 2025 Promo Q8]

Solution	Comments
(a) **[3]** 	
(b) **[3]** 	
(c) **[3]** 	

4.4 Express $\frac{9x+5}{3x+2}$ in the form $a-\frac{b}{3x+2}$ where a and b are positive constants.

Hence, describe a sequence of transformations which transform the graph of $y=\frac{9x+5}{3x+2}$ to the graph of $y=\frac{1}{x}$. [5]

[RVHS 2026 Prelim P1 Q10(b)]

Solution		Comments
[5]		
(b)	$\frac{9x+5}{3x+2}=3-\frac{1}{3x+2} \quad (a=3,\ b=1)$ $y=3-\frac{1}{3x+2} \xrightarrow{y\to y+3} y=-\frac{1}{3x+2}$ $\xrightarrow{y\to -y} y=\frac{1}{3x+2}$ $\xrightarrow{x\to \frac{x}{3}} y=\frac{1}{x+2}$ $\xrightarrow{x\to x-2} y=\frac{1}{x}$ 1. Translate 3 units in the negative y-direction 2. Reflect about the x-axis 3. Scale by factor 3 parallel to the x-axis 4. Translate 2 units in the positive x-direction	

5 Functions

5.1 The function f is defined by

$$\mathrm{f}: x \to \left|\frac{2x+4}{3-x}\right|, \qquad x \in \mathbb{R}, \qquad x \neq 3.$$

(a) Sketch the graph of $y = \mathrm{f}(x)$, giving the equations of any asymptotes and the coordinates of the points where the curve meets the axes. [3]

(b) Hence state the range of f. [1]

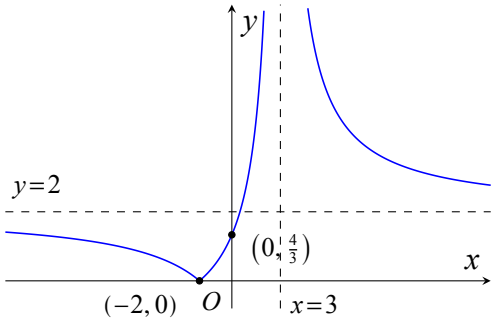
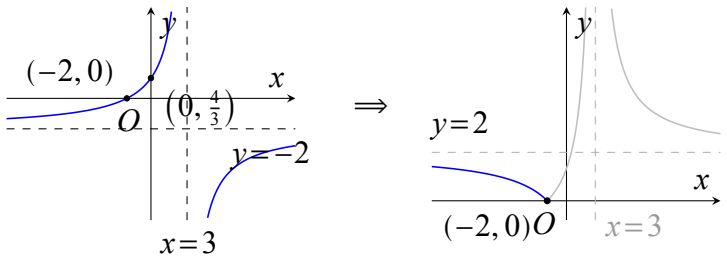
(c) Explain why the function f^2 does not exist. [1]

The domain of f is further restricted to $x \leqslant a$, where a is a constant.

(d) State the greatest value of a such that the function f^{-1} exists. [1]

(e) Hence find $\mathrm{f}^{-1}(x)$ and state its domain. [4]

[2023 A-Level P1 Q7]

Solution		Comments
(a) **[3]**	$y=\left\|\frac{2x+4}{3-x}\right\|=\left\|-2+\frac{10}{3-x}\right\|$ Asymptotes: $y=\|-2\|=2, \quad x=3$ $x=0: \quad y=\left\|\frac{4}{3}\right\|=\frac{4}{3}$ $y=0: \quad x=-2$  	
(b) **[1]**	$\mathrm{R_f}=[0,\infty)$	
(c) **[1]**	$\mathrm{R_f}=[0,\infty), \quad \mathrm{D_f}=(-\infty,3)\cup(3,\infty)$ $3\in\mathrm{R_f},\ 3\notin\mathrm{D_f} \Rightarrow \mathrm{R_f}\not\subseteq\mathrm{D_f}$ $\therefore$ f^2 does not exist	
(d) **[1]**	$a=-2$	
(e) **[4]**	$x\leqslant -2: \quad \mathrm{f}(x)=\left\|\frac{2x+4}{3-x}\right\|=-\frac{2x+4}{3-x}$  $y=\mathrm{f}(x)$ $-3y+xy=2x+4$ $x(y-2)=4+3y$ $x=\frac{4+3y}{y-2}$ $\mathrm{f}^{-1}(x)=\frac{4+3x}{x-2}, \qquad \mathrm{D_{f^{-1}}}=[0,2)$	

5.2 The function f is defined by

$$\mathrm{f}: x \mapsto \frac{ax-3}{5x-4}, \text{ for } x \in \mathbb{R}, \ x \neq \frac{4}{5}, \text{ and } a \in \mathbb{R}.$$

(a) Find the range of f in terms of a. [1]

The function f is such that $\mathrm{f}(x) = \mathrm{f}^{-1}(x)$ for all x in the domain of f.

(b) Find the value of a. [2]

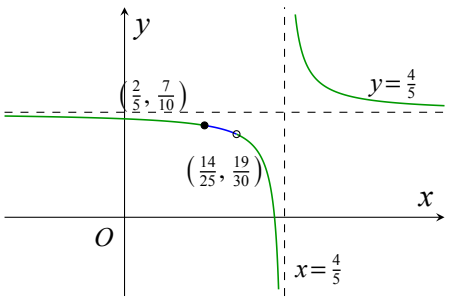
(c) Hence find $\mathrm{f}^{2025}(2)$. [2]

Another function g is defined by

$$\mathrm{g}: x \mapsto \left(x - \frac{1}{5}\right)^2 + \frac{2}{5}, \text{ for } x \in \mathbb{R}, \ 0 < x < \frac{3}{5}.$$

(d) Find the range of fg. [2]

[HCI 2025 Prelim P1 Q8]

Solution		Comments
(a) **[1]**	Horizontal asymptote: $y=\frac{a}{5}$ $\mathrm{R_f}=\mathbb{R}\setminus\left\{\frac{a}{5}\right\}$	
(b) **[2]**	$y=\frac{ax-3}{5x-4}$ $5xy-4y=ax-3$ $5xy-ax=4y-3$ $x=\frac{4y-3}{5y-a}$ $\mathrm{f}^{-1}(x)=\frac{4x-3}{5x-a},\quad x\in\mathbb{R},\ x\neq\frac{a}{5}$ $\mathrm{f}(x)=\mathrm{f}^{-1}(x)\ \Rightarrow\ a=4$	
(c) **[2]**	$\mathrm{f}^{-1}(x)=\mathrm{f}(x)\ \Rightarrow\ \mathrm{f}^2(x)=\mathrm{ff}^{-1}(x)=x$ $\mathrm{f}^{2025}(2)=\mathrm{f}^2\left(\mathrm{f}^{2023}(2)\right)$ $=\mathrm{f}^2\left(\mathrm{f}^{2021}(2)\right)$ $=\cdots$ $=\mathrm{f}(2)$ $=\frac{4(2)-3}{5(2)-4}=\frac{5}{6}$	
(d) **[2]**	$\mathrm{R_g}=\left[\frac{2}{5},\frac{14}{25}\right)$ $\mathrm{f}(x)=\frac{4x-3}{5x-4}$ $\left(0,\frac{3}{5}\right)\xrightarrow{\mathrm{g}}\left[\frac{2}{5},\frac{14}{25}\right)\xrightarrow{\mathrm{f}}\left(\frac{19}{30},\frac{7}{10}\right]$ $\mathrm{R_{fg}}=\left(\frac{19}{30},\frac{7}{10}\right]$  	

5.3 The function f is given by

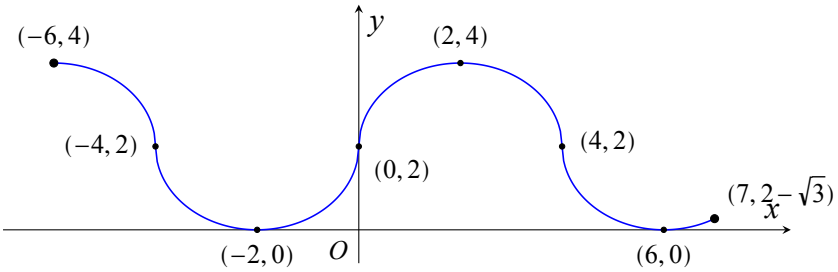
$$\mathrm{f}(x)=\begin{cases}2+\sqrt{2^2-(x-2)^2}, & 0<x\leqslant 4,\\ 2-\sqrt{2^2-(x-6)^2}, & 4<x\leqslant 8.\end{cases}$$

It is given that $\mathrm{f}(x)=\mathrm{f}(x+8)$ for all real values of x.

(a) On the diagram in the Printed Answer Book, sketch the graph of $y=\mathrm{f}(x)$ for $-6\leqslant x\leqslant 7$, indicating clearly the coordinates of the end points and the points where the graph cuts the axes. [3]

(b) Without integrating, write down the exact area of the region bounded by $y=\mathrm{f}(x)$, the line $x=4$, the x-axis and the y-axis. [1]

[VJC 2025 Prelim P1 Q5(a), (b)]

Solution		Comments
(a) **[3]**	$y=2+\sqrt{2^2-(x-2)^2}$ $y-2=\sqrt{2^2-(x-2)^2}$ $(x-2)^2+(y-2)^2=2^2$ Centre $(2,2)$, radius 2. Since it is a positive square root, it is the top half of this circle. $y=2-\sqrt{2^2-(x-6)^2}$ $y-2=-\sqrt{2^2-(x-6)^2}$ $(x-6)^2+(y-2)^2=2^2$ Centre $(6,2)$, radius 2. Since it is a negative square root, it is the bottom half of this circle.  	
(b) **[1]**	$\text{Area}=2\times 4+\frac{1}{2}\pi(2^2)=8+2\pi$	

5.4 The function f is defined by

$$\mathrm{f}(x) = 1 - ax,\ x \in \mathbb{R},\ \text{where } a \text{ is a real constant.}$$

The function g is defined by

$$\mathrm{g}(x) = \begin{cases} x^2 + 3 & \text{for} \quad 0 \leqslant x \leqslant 2, \\ 7 - x & \text{for} \quad 2 < x \leqslant 7. \end{cases}$$

(i) Find the set of possible values of a such that f^{-1} exist. [1]

(ii) If $a = 2$, describe a sequence of transformations that transform the graph of $y = \mathrm{g}(x)$ onto the graph of $y = \mathrm{fg}(x)$. [3]

(iii) The function gg is defined by

$$\mathrm{gg}(x) = \begin{cases} \mathrm{h}_1(x) & \text{for} \quad 0 \leqslant x \leqslant 2, \\ \mathrm{h}_2(x) & \text{for} \quad 2 < x < 5, \\ \mathrm{h}_3(x) & \text{for} \quad 5 \leqslant x \leqslant 7. \end{cases}$$

Find $\mathrm{h}_1(x)$, $\mathrm{h}_2(x)$ and $\mathrm{h}_3(x)$. [3]

[ACJC 2022 Prelim P2 Q1]

Solution		Comments
(i) **[1]**	$a \in \mathbb{R} \setminus \{0\}$	
(ii) **[3]**	$y = \mathrm{fg}(x) = 1 - 2\mathrm{g}(x)$ Reflection of the graph of $y = \mathrm{g}(x)$ in the x-axis, followed by scaling by scale factor 2 parallel to the y-axis, followed by translation 1 unit in the positive y-direction. **OR** Translation of the graph of $y = \mathrm{g}(x)$ by $\frac{1}{2}$ unit in the negative y-direction followed by reflection in the x-axis followed by scaling by scale factor 2 parallel to the y-axis. **OR** Translation of the graph of $y = \mathrm{g}(x)$ by $\frac{1}{2}$ unit in the negative y-direction followed by scaling by scale factor 2 parallel to the y-axis followed by reflection in the x-axis.	
(iii) **[3]**	$\mathrm{gg}(x) = \begin{cases} 7-(x^2+3) & \text{for} \quad 0 \leqslant x \leqslant 2 \\ 7-(7-x) & \text{for} \quad 2 < x < 5 \\ (7-x)^2+3 & \text{for} \quad 5 \leqslant x \leqslant 7 \end{cases}$ $\therefore \mathrm{gg}(x) = \begin{cases} 4-x^2 & \text{for} \quad 0 \leqslant x \leqslant 2 \\ x & \text{for} \quad 2 < x < 5 \\ x^2-14x+52 & \text{for} \quad 5 \leqslant x \leqslant 7 \end{cases}$	

5.5 Functions f and g are defined by

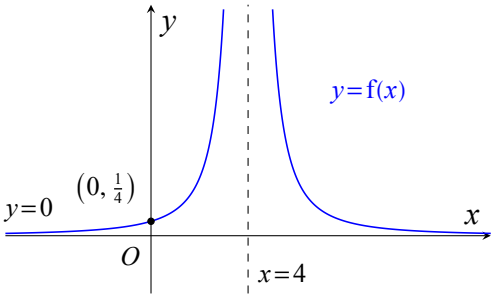
$$\mathrm{f}: x \to \frac{4}{(x-4)^2}, \quad x \in \mathbb{R},\ x \neq 4,$$

$$\mathrm{g}: x \to \ln\left(1+\frac{1}{x}\right), \quad x \in \mathbb{R},\ x > 0.$$

(a) Sketch the graph of $y = \mathrm{f}(x)$, stating the equations of any asymptotes, the coordinates of the points where it crosses the axes and the coordinates of the turning points, if any. [2]

(b) Show that gf exists. Hence find the rule, domain and range of gf. [4]

[CJC 2025 Prelim P2 Q2(a), (b)]

Solution		Comments
(a) **[2]**		

(b)
[4]

For gf to exist, $R_f \subseteq D_g$,

$R_f = (0, \infty)$

$D_g = (0, \infty)$

Since $R_f \subseteq D_g$, gf exists (shown)

$$\begin{aligned} \text{gf}(x) &= \text{g}\left[\frac{4}{(x-4)^2}\right] \\ &= \ln\left[1+\left(\frac{1}{\frac{4}{(x-4)^2}}\right)\right] \\ &= \ln\left[1+\frac{(x-4)^2}{4}\right] \end{aligned}$$

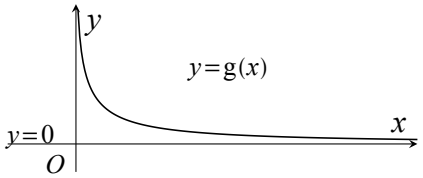
$D_{gf} = D_f = (-\infty, 4) \cup (4, \infty)$

OR

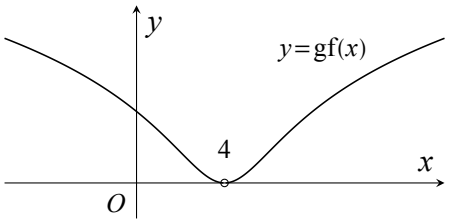
$D_{gf} = D_f = \mathbb{R} \setminus \{4\}$

Method 1: Mapping

$D_f = (-\infty, 4) \cup (4, \infty) \xrightarrow{f} R_f = (0, \infty) \xrightarrow{g} R_{gf} = (0, \infty)$



Method 2: Sketch $y = \text{gf}(x)$ for given domain



$R_{gf} = (0, \infty)$

5.6 The function g is defined by $g(0)=g(1)=1$, $g(2n)=2+g(n)$ and $g(2n+1)=1+g(2n)$, for integer n.

(i) Find $g(4)$, $g(7)$ and $g(12)$.

(ii) Does g have an inverse? Justify your answer. [3]

[2016 A-Level P1 Q10(b)]

Solution		**Comments**
(i)	$g(4)=2+g(2)$ $=2+2+g(1)$ $=2+2+1+g(0)$ $=2+2+1+1$ $=6$ $g(7)=1+g(6)$ $=1+2+g(3)$ $=1+2+1+g(2)$ $=1+2+1+2+1+1$ $=8$ $g(12)=2+g(6)$ $=2+2+1+2+1+1$ $=9$	
(ii) **[3]**	$g(8)=2+g(4)$ $=2+6$ $=8$ $=g(7)$ Since $g(8)=g(7)=8$, g is not one-one, so g^{-1} does not exist.	

6 Sequences and Series

6.1 It is given $\sum_{r=1}^{n} r^3 = \frac{n^2(n+1)^2}{4}$ and $u_r = 2r^3 + 5$.

(i) Find $\sum_{r=1}^{n} u_r$. [2]

(ii) Hence, or otherwise, find $\sum_{r=2}^{n} (2(r+2)^3 + 5)$. [3]

(iii) Explain why the series $\sum_{r=1}^{\infty} u_r$ does not converge. [1]

[JPJC 2025 Prelim P2 Q8(b)]

Solution		Comments
(i) **[2]**	$\sum_{r=1}^{n}(2r^3+5) = 2\sum_{r=1}^{n} r^3 + \sum_{r=1}^{n} 5$ $= 2\left(\frac{n^2(n+1)^2}{4}\right) + 5n$ $= \frac{n^2(n+1)^2}{2} + 5n$	
(ii) **[3]**	$\sum_{r=2}^{n}(2(r+2)^3+5) = (2(2+2)^3+5) + \cdots + (2(n+2)^3+5)$ $= \sum_{r=4}^{n+2}(2r^3+5)$ $= \sum_{r=1}^{n+2}(2r^3+5) - \sum_{r=1}^{3}(2r^3+5)$ $= \frac{(n+2)^2(n+3)^2}{2} + 5(n+2) - \left[\left(\frac{3^2(4)^2}{2}\right) + 5(3)\right]$ $= \frac{(n+2)^2(n+3)^2}{2} + 5n - 77$	
(iii) **[1]**	From (i) $\sum_{r=1}^{n} u_r = \frac{n^2(n+1)^2}{2} + 5n$ As $n \to \infty$, $\frac{n^2(n+1)^2}{2} \to \infty$, $5n \to \infty$ Hence, $\sum_{r=1}^{n} u_r = \frac{n^2(n+1)^2}{2} + 5n \to \infty$, series $\sum_{r=1}^{\infty} u_r$ does not converge.	

6.2 It is given that $\sum_{r=2}^{n} \frac{1}{r(r^2-1)} = \frac{1}{4} + \frac{1}{2(n+1)} - \frac{1}{2n}$.

(a) Show that $\sum_{r=2}^{n} \frac{1}{r(r^2-1)}$ is less than $\frac{1}{4}$. [2]

(b) Give a reason why the series $\sum_{r=2}^{\infty} \frac{1}{r(r^2-1)}$ converges, and write down its value. [2]

(d) Find $\sum_{r=m+1}^{N} \frac{1}{(r-m)(r-m+1)(r-m+2)}$, where m and N are integers with $N > m > 0$.
(There is no need to express your answer as a single algebraic fraction.) [2]

[VJC 2025 Prelim P1 Q6(a), (b), (d)]

Solution		Comments
(a) **[2]**	$\frac{1}{2(n+1)} - \frac{1}{2n} = \frac{n-(n+1)}{2n(n+1)} = \frac{-1}{2n(n+1)} < 0$ for all $n \geqslant 2$ $\sum_{r=2}^{n} \frac{1}{r(r^2-1)} = \frac{1}{4} + \frac{1}{2(n+1)} - \frac{1}{2n} < \frac{1}{4}$	
(b) **[2]**	As $n \to \infty$, $\frac{1}{2(n+1)} \to 0$ and $\frac{1}{2n} \to 0$, $\therefore \sum_{r=2}^{n} \frac{1}{r(r^2-1)} \to \frac{1}{4}$, a constant. Hence $\sum_{r=2}^{\infty} \frac{1}{r(r^2-1)}$ converges and $\sum_{r=2}^{\infty} \frac{1}{r(r^2-1)} = \frac{1}{4}$.	
(d) **[2]**	$\sum_{r=m+1}^{N} \frac{1}{(r-m)(r-m+1)(r-m+2)}$ $= \frac{1}{1(2)(3)} + \frac{1}{2(3)(4)} + \cdots + \frac{1}{(N-m)(N-m+1)(N-m+2)}$ $= \sum_{r=2}^{N-m+1} \frac{1}{(r-1)(r)(r+1)}$ $= \sum_{r=2}^{N-m+1} \frac{1}{r(r^2-1)}$ $= \frac{1}{4} + \frac{1}{2(N-m+2)} - \frac{1}{2(N-m+1)}$	

6.3 **(a)** It is given that $\sum_{r=1}^{n} \frac{2}{r(r+1)(r+2)} = \frac{1}{2} - \frac{1}{(n+1)(n+2)}$.

(i) State the sum to infinity of the series. [1]

(ii) Find the sum of the first n terms of the series $\frac{2}{17\times18\times19} + \frac{2}{18\times19\times20} + \frac{2}{19\times20\times21} + \cdots$. [2]

(iii) Find $\sum_{r=2}^{n} \frac{1}{r(r^2-1)}$. [3]

(b) The sequence $u_1, u_2, u_3, \ldots$ is defined by $u_1 = 3$, $u_{n+1} = 2 - \frac{4}{u_n}$, $n \geqslant 1$.

(i) Find the values of u_2, u_3, u_4 and u_{100}. [2]

(ii) Hence find $\sum_{r=1}^{3n-2} u_r$ in terms of n, simplifying your answer. [2]

[RI 2025 Promo Q9]

Solution		Comments
(a)(i) **[1]**	The sum to infinity is $\frac{1}{2}$.	
(a)(ii) **[2]**	$\sum_{r=17}^{n+16} \frac{2}{r(r+1)(r+2)}$ $= \sum_{r=1}^{n+16} \frac{2}{r(r+1)(r+2)} - \sum_{r=1}^{16} \frac{2}{r(r+1)(r+2)}$ $= \frac{1}{2} - \frac{1}{(n+17)(n+18)} - \left(\frac{1}{2} - \frac{1}{17\times 18}\right)$ $= \frac{1}{306} - \frac{1}{(n+17)(n+18)}$	
(a)(iii) **[3]**	$\sum_{r=2}^{n} \frac{1}{r(r^2-1)}$ $= \frac{1}{2}\sum_{r=2}^{n} \frac{2}{(r-1)r(r+1)}$ $= \frac{1}{2}\left(\frac{2}{1\times 2\times 3} + \cdots + \frac{2}{(n-1)n(n+1)}\right)$ $= \frac{1}{4} - \frac{1}{2n(n+1)}$	
(b)(i) **[2]**	$u_2 = 2 - \frac{4}{3} = \frac{2}{3}$ $u_3 = 2 - \frac{4}{\frac{2}{3}} = -4$ $u_4 = 2 - \frac{4}{-4} = 3$ $u_{100} = u_{97} = u_{94} = \cdots = u_1 = 3$ Or use GC to obtain the values. [GC SCREEN PLACEHOLDER] GC 2: table of u_n, $u_{n+1} = 2 - \frac{4}{u_n}$, $u_1 = 3$, $n = 1$ to 7	
(b)(ii) **[2]**	$\sum_{r=1}^{3n-2} u_r = n\left(3 + \frac{2}{3} - 4\right) - \frac{2}{3} + 4 = \frac{10-n}{3}$	

6.4 A sequence is such that $u_{n+1} = u_n + 1.5$, for $n = 1, 2, 3, \ldots$.

(a) Given that $u_{50} = 99u_1$, show that $u_1 = 0.75$. [2]

[NYJC 2025 Prelim P1 Q8(a)]

Solution		Comments
(a) **[2]**	Note that the sequence is an arithmetic progression with common difference 1.5 $u_{50} = u_1 + 49(1.5)$ $99u_1 = u_1 + 73.5$ $u_1 = 0.75$	

6.5 A sequence $u_1, u_2, u_3, \ldots$ is defined by

$$u_{n+1} = 2u_n\,(1-u_n)$$

for all positive integers n.

(a) Find the possible limits of the sequence, if the sequence is convergent. [2]

(b) For each of the following, evaluate u_2 and u_3, and state whether the sequence is convergent:

- $u_1 = 0$,
- $u_1 = 0.01$,
- $u_1 = -0.01$. [6]

[EJC 2025 Prelim P2 Q2]

Solution		Comments
(a) **[2]**	Let L be the limit of the sequence $u_{n+1} = 2u_n\,(1-u_n)$. $L = 2L(1-L)$ $2L^2 - L = 0$ $L(2L-1) = 0$ $L = 0$ or $L = \frac{1}{2}$	
(b) **[6]**	When $u_1 = 0$, $u_2 = 0$, $u_3 = 0$, sequence is convergent. When $u_1 = 0.01$, $u_2 = 0.0198$ (exact), $u_3 = 0.0388$ (3 s.f.), sequence is convergent. When $u_1 = -0.01$, $u_2 = -0.0202$ (exact), $u_3 = -0.0412$ (3 s.f.), sequence is not convergent.	